Kushal Sachdeva <kushals@vasantvalley.edu.in>

Thanks for filling out this form: RoboCupJunior Soccer TDP form

Forms Response Receipts <forms-receipts-noreply@google.com> Sun, Jun 28, 2026 at 7:28 PM
To: kushals@vasantvalley.edu.in

Thanks for filling out this form: RoboCupJunior Soccer TDP form

Here's what was received.

Edit response

RoboCupJunior Soccer TDP form

Welcome to the RoboCupJunior Soccer Technical Documentation Form!

This form will guide your team through writing a clear, complete report of your robot and your work.

1. **Who fills this out?**

   • Every team member must help—your work here counts toward your competition score.

   • If a section doesn't apply to your robot, just write "N/A."

2. **Why does it matter?**

   • It help us to do more efficient interviews

   • Future teams will learn from your design and ideas.

3.

Tips for success:

• Keep notes, photos, and sketches as you build during the year—they make writing this form much faster.

• Write in short, clear sentences—focus on the key facts. There is a character limit on any question, so use your space wisely.

4.

What happens next?

• Your completed form is private until competition time.

• After the event, it will be published alongside your poster.

Helpful links
• Past years' reports: https://github.com/robocup-junior/awesome-rcj-soccer
• RoboCupJunior Discord: https://discord.gg/eA4fwnN5
• RoboCupJunior Forum: https://junior.forum.robocup.org/
• Scoring rubrics (read before you start): https://robocup-junior.github.io/soccer-rules/master/scoring.html

Your email (**kushals@vasantvalley.edu.in**) was recorded when you submitted this form.

For any question while filing this form or in general about RCJ use our Discord server or the RoboCupJunior Forum

RCJ Discord

TDP File Upload (Not required)

If you have a longer TDP that you want to share with the other teams, you can upload it here

No files submitted

What is your team's name? *

VVS Ballers

What league do you participate in? *

- [ ] Vision League
- [x] IR League

Where are you from?
*

India

If other teams have questions about your robot, now or in the future, what email address(es) can we publish along with this document for people to reach you?

(You can put in multiple email addresses, like multiple team members, an email for the whole team or both. Feel free to share other ways of communication like Discord handles)
*

Team email ID - vvsballers@gmail.com
Kushal Sachdeva - kushal.sach78@gmail.com

Team Social Media Links (if you have any)

www.instagram.com/vvs_ballers

Upload a photo of your whole team with your mentor and robots

Note: This is not mandatory and will be published along with your TDP if you choose to upload something

Submitted files

IMG_5779 - Kushal Sachdeva.jpg

What are the names of the team members and their role(s)?
*

example:
John Doe: Circuit and PCB design
Jane Doe: Programming
...

Kushal Sachdeva (Team Leader) — Electrical, Software & CAD Engineer
Darsh Goel — Mechanical & CAD Engineer

How often did your team meet?
(e.g. 90 minutes once per week or a day every weekend.)
*

8–10 hours/day from 7–27 June; 4 hours/day from February to May

Where did you meet to work on your robot?
(e.g. a robotics room at school, at some other place, one of your homes, school library etc.) *

A robotics room set up at one of our homes

When did your team start working on this year's robot? *

February 2026

Which RoboCupJunior competitions have you competed in and in which leagues?
*

Example:
German Open 2023: 1v1 Soccer Entry Standard Kit

German Open 2024: Lightweight League
European Championship 2024: Lightweight League
...

2023–24 Season
- RoboCup Regionals – Best Innovation Award (qualified to Nationals)
- RoboCup Nationals – Best Robot Design Award (qualified to Asia-Pacific)

2024–25 Season
- RoboCup Regionals – 3rd Place (qualified to Nationals)
- RoboCup Nationals – 3rd Place (qualified to Internationals)
- RoboCup Internationals, Brazil – Top 20 worldwide

2025-26 Season
- RoboCup Regionals – Best Innovation Award (qualified to Nationals)
- RoboCup Nationals – 1st Place (qualified to Internationals)

Which parts of your work received the most contribution from your mentor?

*

Example:
We couldn't manage to get our I2C compass to work, so our mentor helped us fixing the code

External Nano caps motor-driver current (overheating); bulk cap stopped Ultrasonic PCB resets; IR smoothing + camera fixed IR's short range.

How did you manage the workload? *

Tools you used to break the work down, assign the work and communicate.
Example:

We communicated through a WhatsApp group and assigned the tasks using monday.com. We also used GitHub for issues and code.

We planned in Notion (every task assigned per member with a deadline), shared all PCB, 3D-design and code files on Google Drive, and ran a WhatsApp group for daily task lists and progress updates.

Which AI tools did you use? *

It's not only fine to use AI for some of your work, it is also recommended to learn to use it well.
Think of things like your code, your poster, your mechanical and electronic designs to remember what you used.

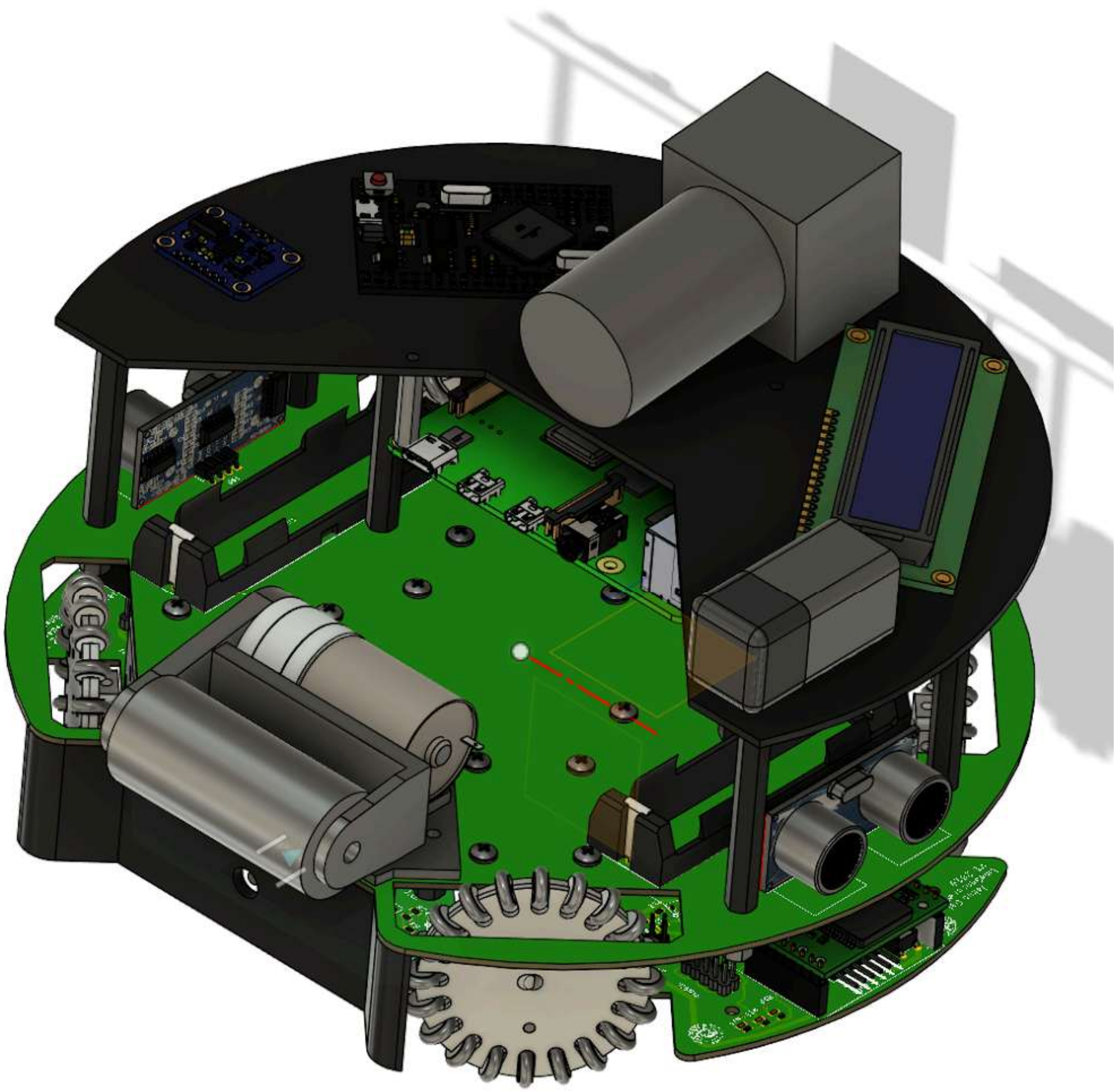
We used Claude's Fusion 360 MCP to help design the supporting beams that hold the IR sensor, and to help lay out the Power PCB and Ultrasonic PCB

Robot 1

The photos of the first robot

Robot 1 Overall View *

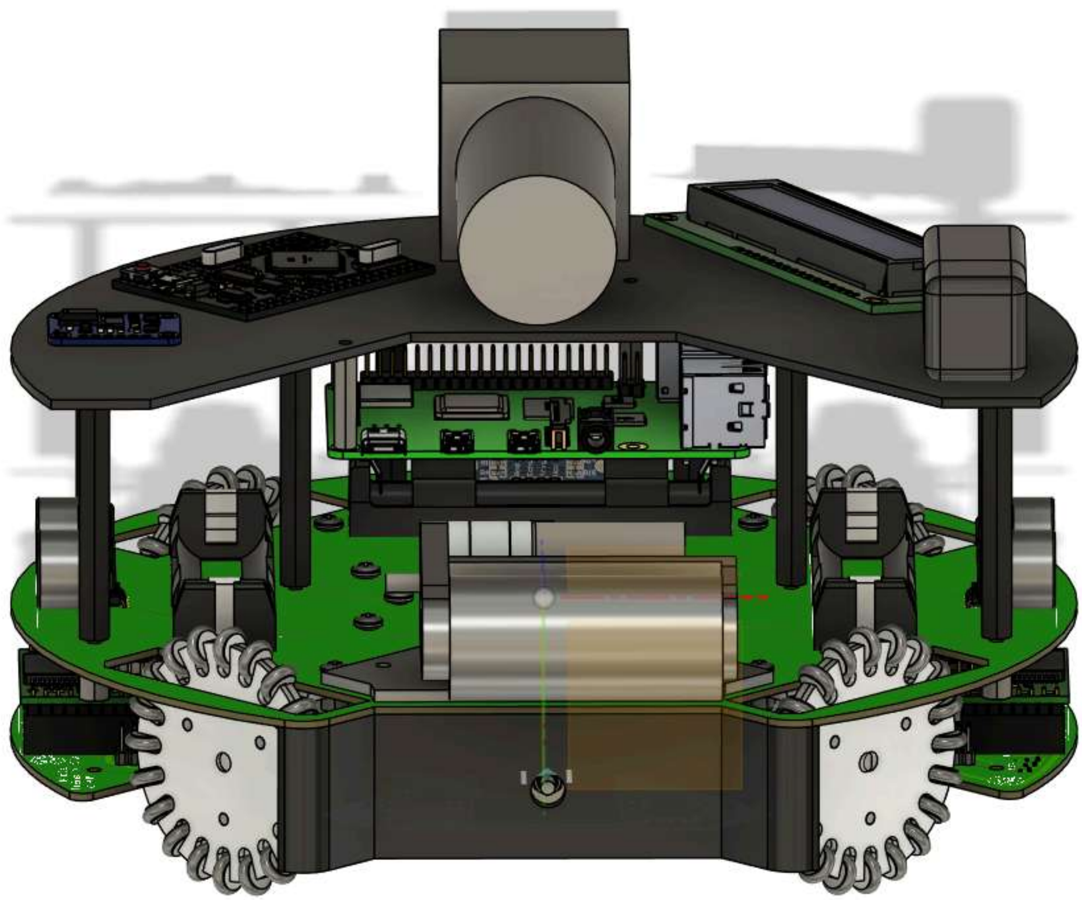
Your entire robot in one photo



Submitted files

Robot 1 Overall View - Kushal Sachdeva.HEIC

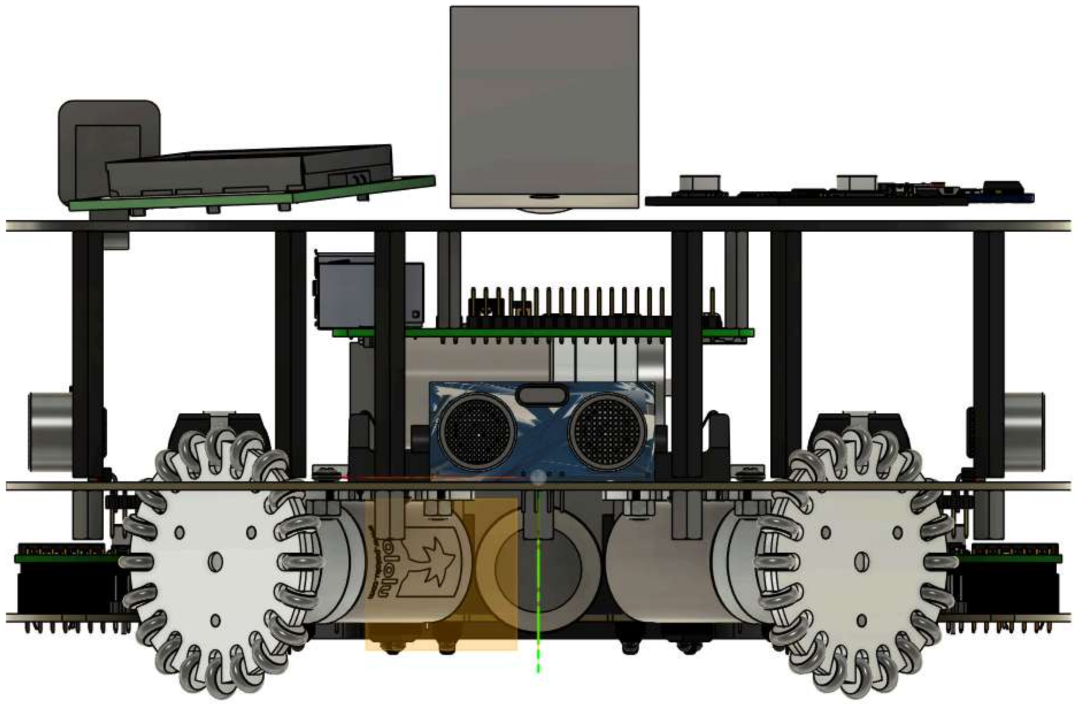
Robot 1 Front view *



Submitted files

Robot 1 Front view - Kushal Sachdeva.HEIC

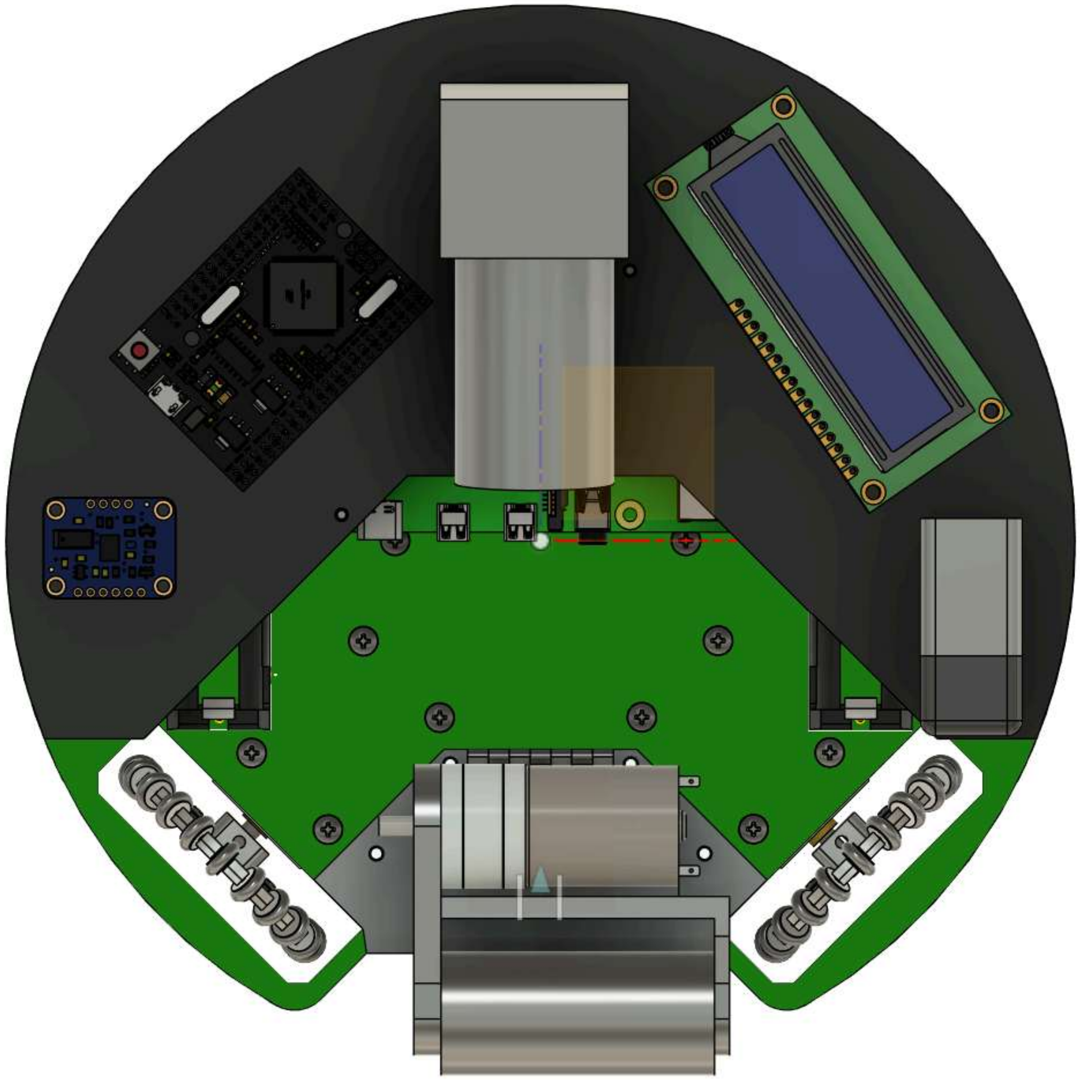
Robot 1 Back view *



Submitted files

Robot 1 Back view - Kushal Sachdeva.HEIC

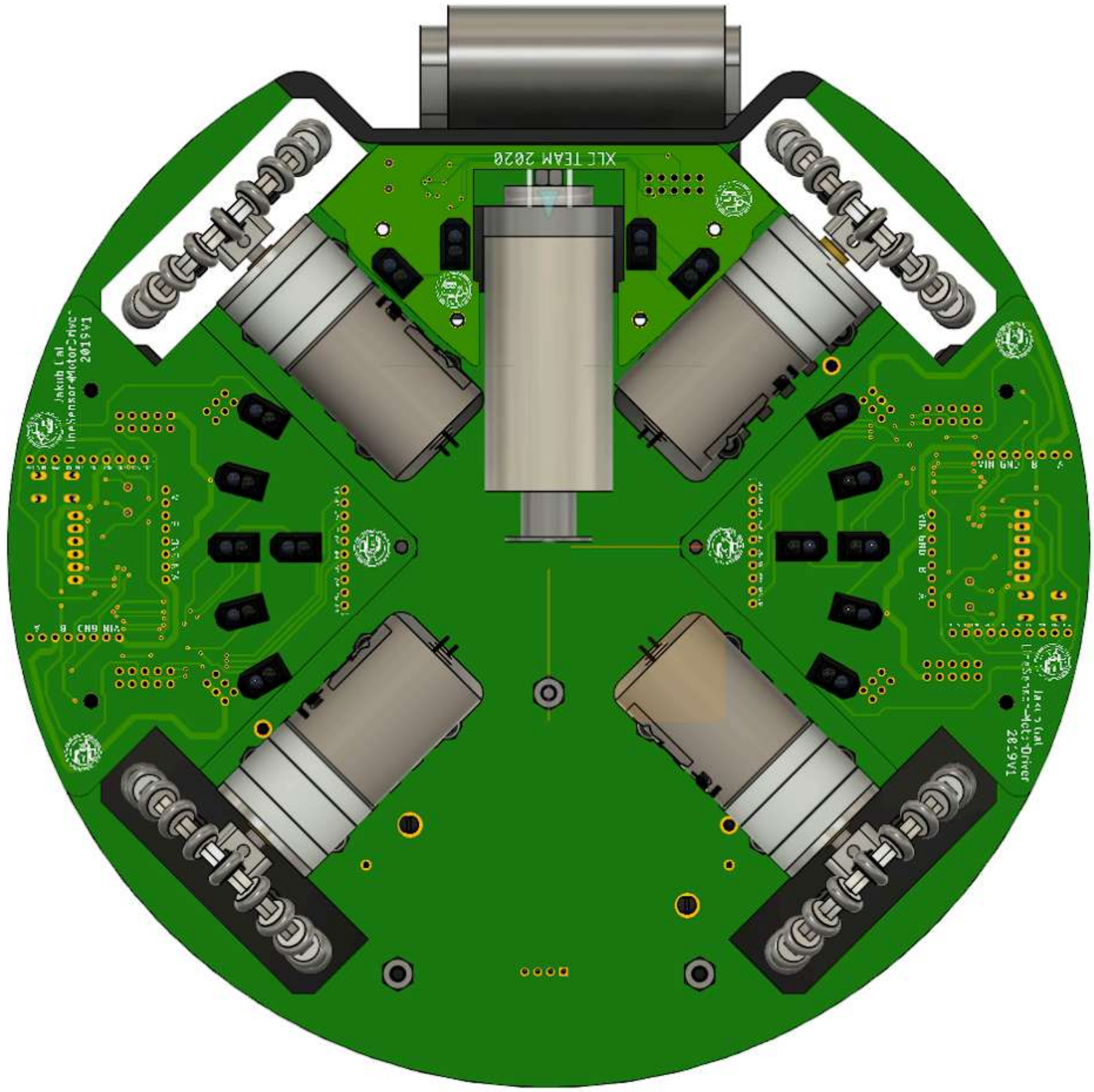
Robot 1 Top View *



Submitted files

Robot 1 Top View - Kushal Sachdeva.HEIC

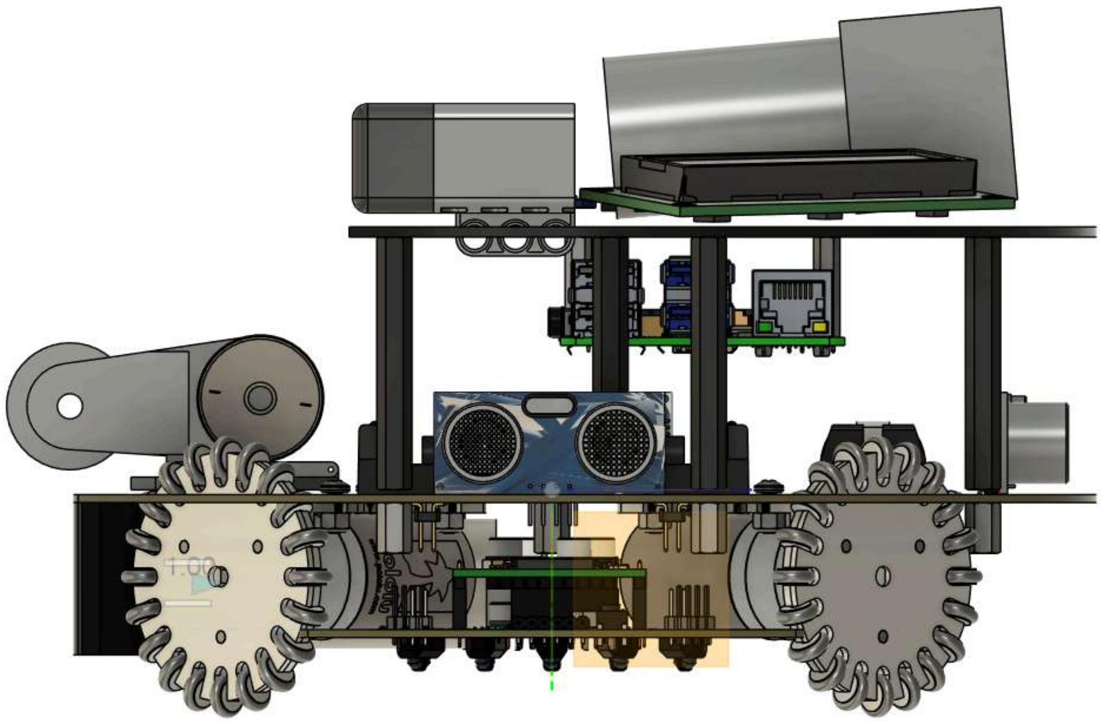
Robot 1 Bottom View *



Submitted files

Robot 1 Bottom View - Kushal Sachdeva.HEIC

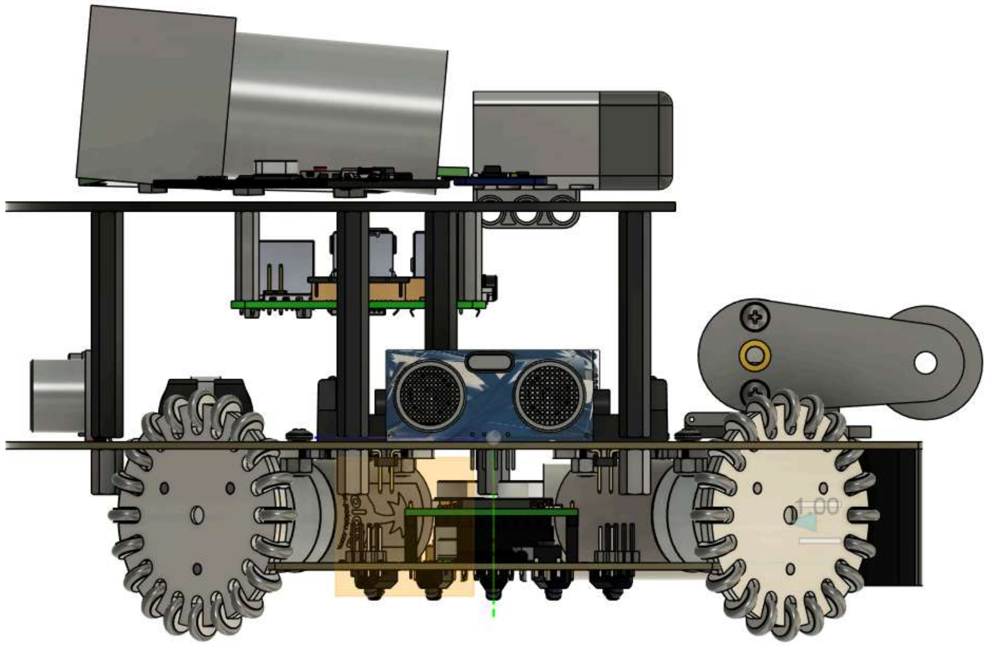
Robot 1 Right View *



Submitted files

Robot 1 Right View - Kushal Sachdeva.HEIC

Robot 1 Left View *



Submitted files

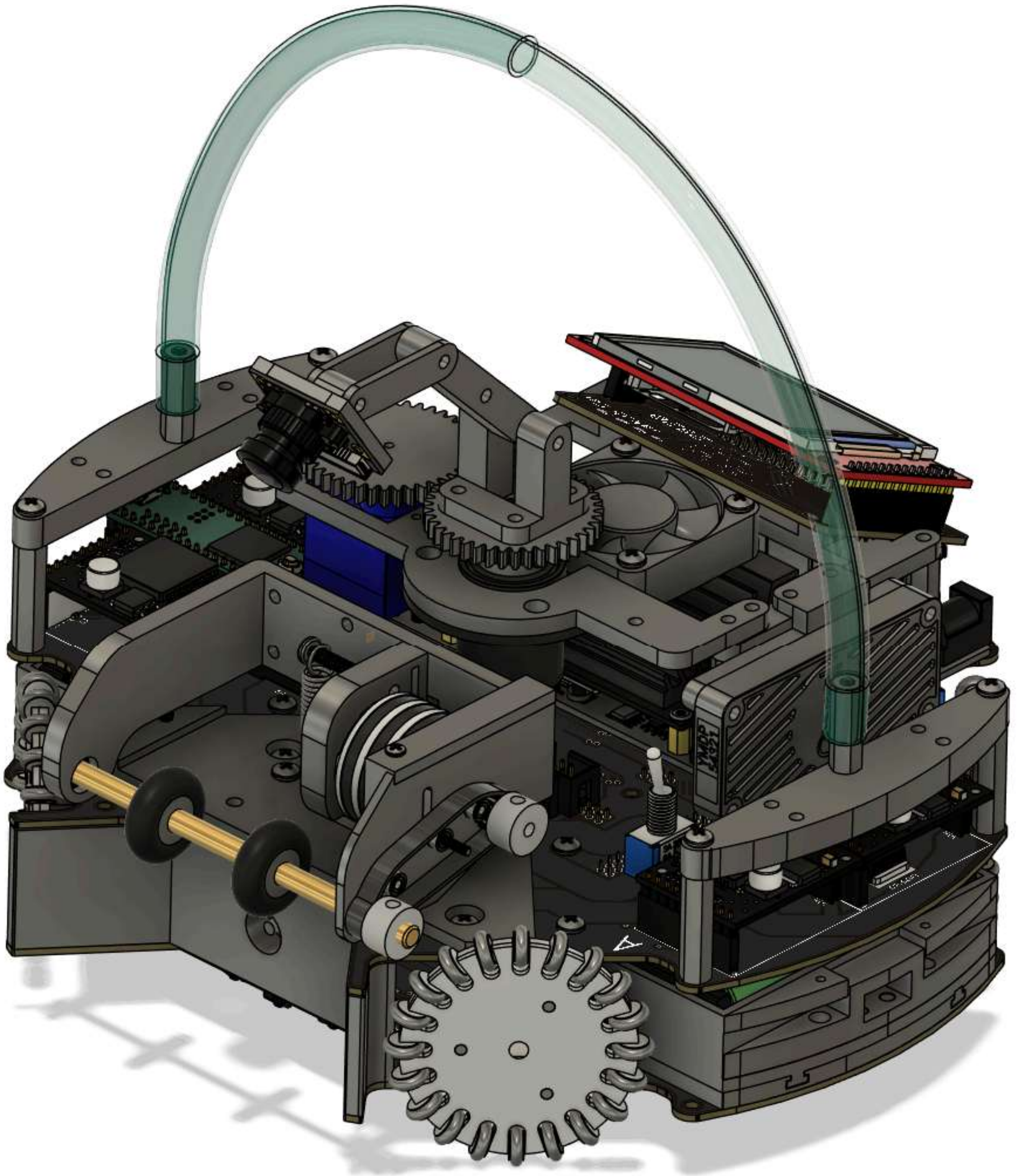
Robot 1 Left View - Kushal Sachdeva.HEIC

Robot 2

The photos of the second robot

Robot 2 Overall View *

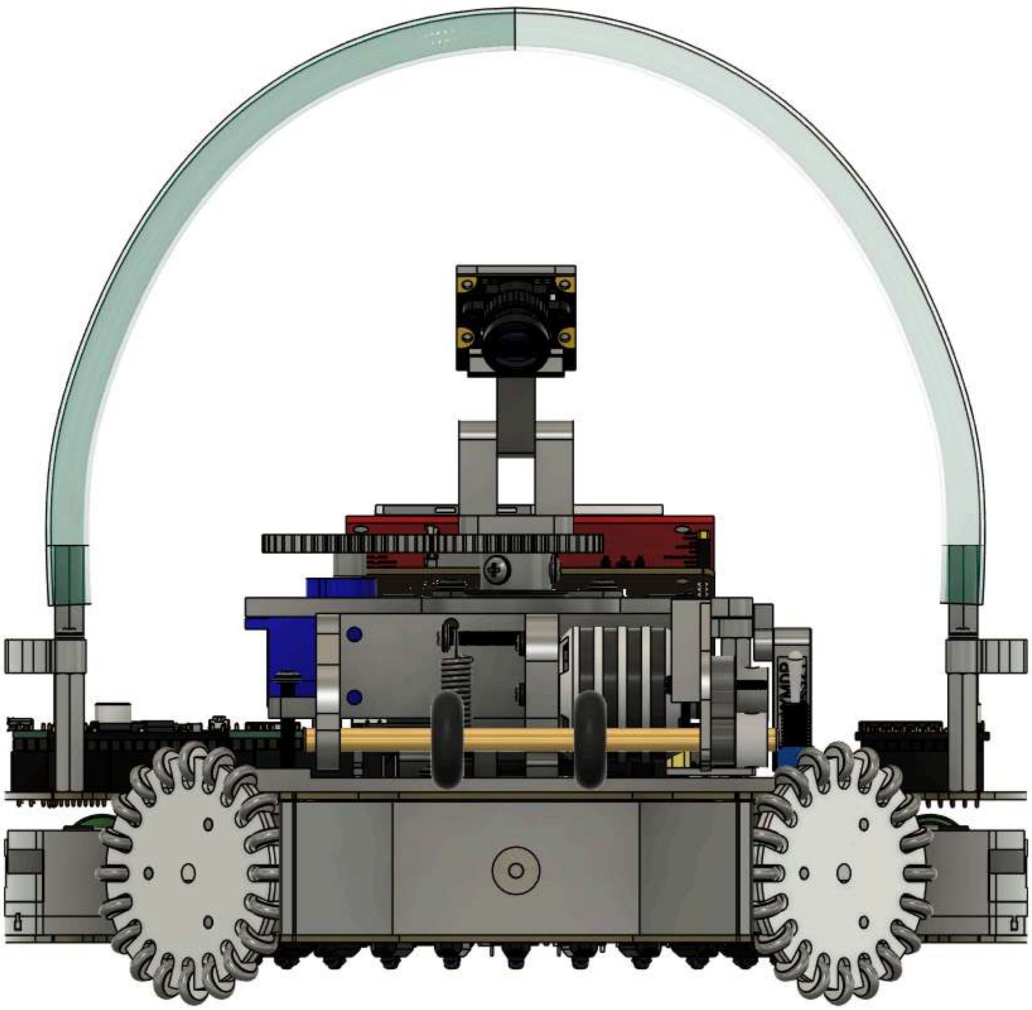
Your entire robot in one photo



Submitted files

Robot 2 Overall View - Kushal Sachdeva.HEIC

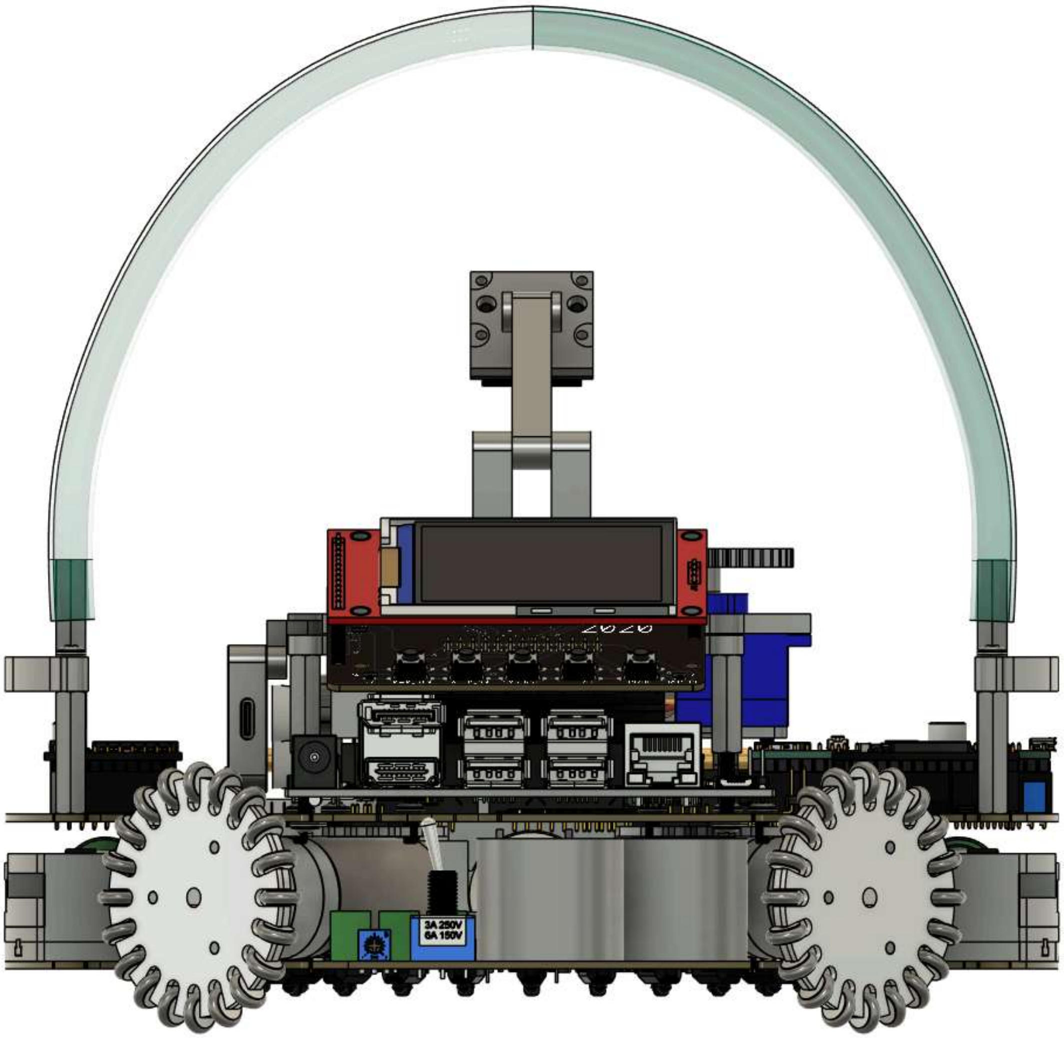
Robot 2 Front view *



Submitted files

Robot 2 Front view - Kushal Sachdeva.HEIC

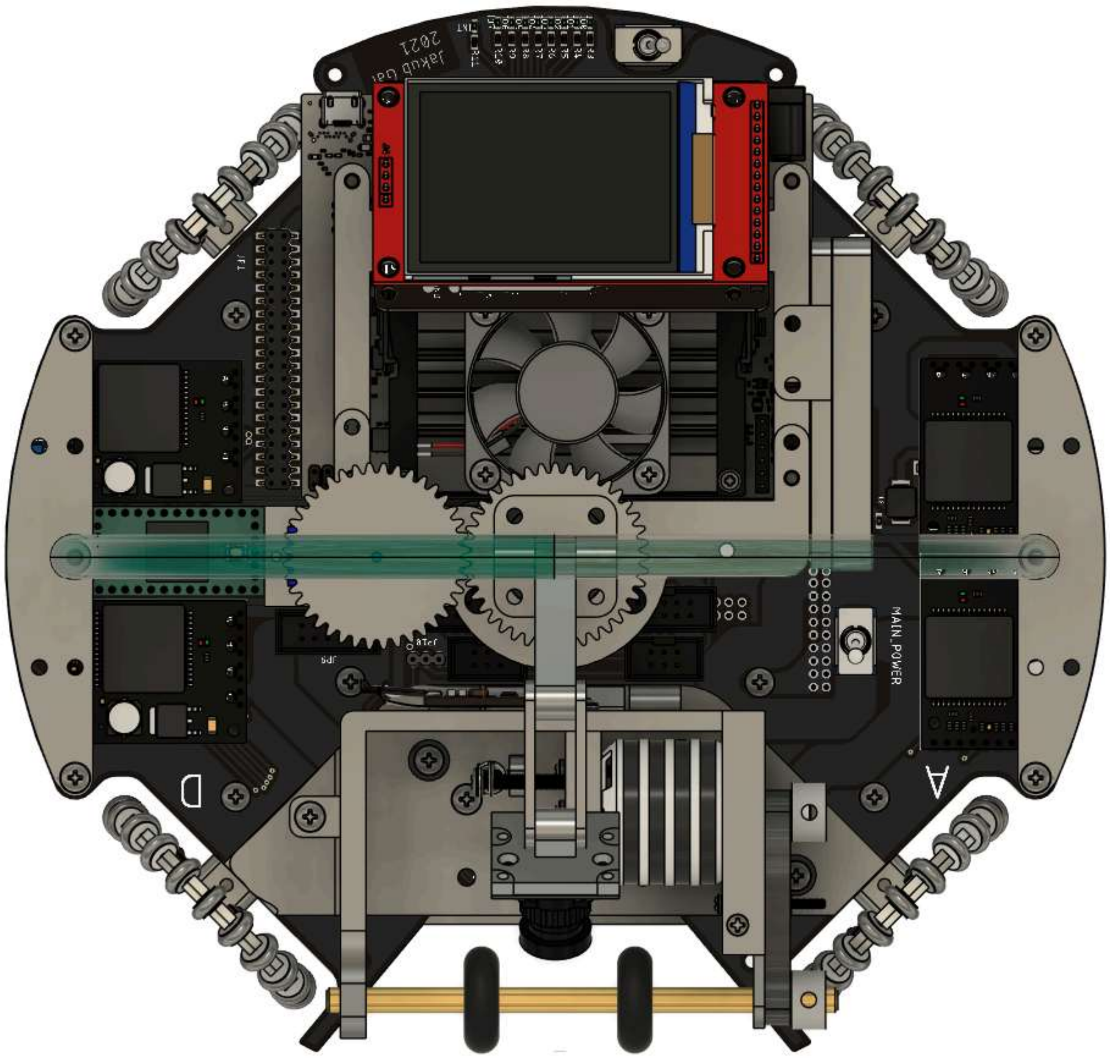
Robot 2 Back view *



Submitted files

Robot 2 Back view - Kushal Sachdeva.HEIC

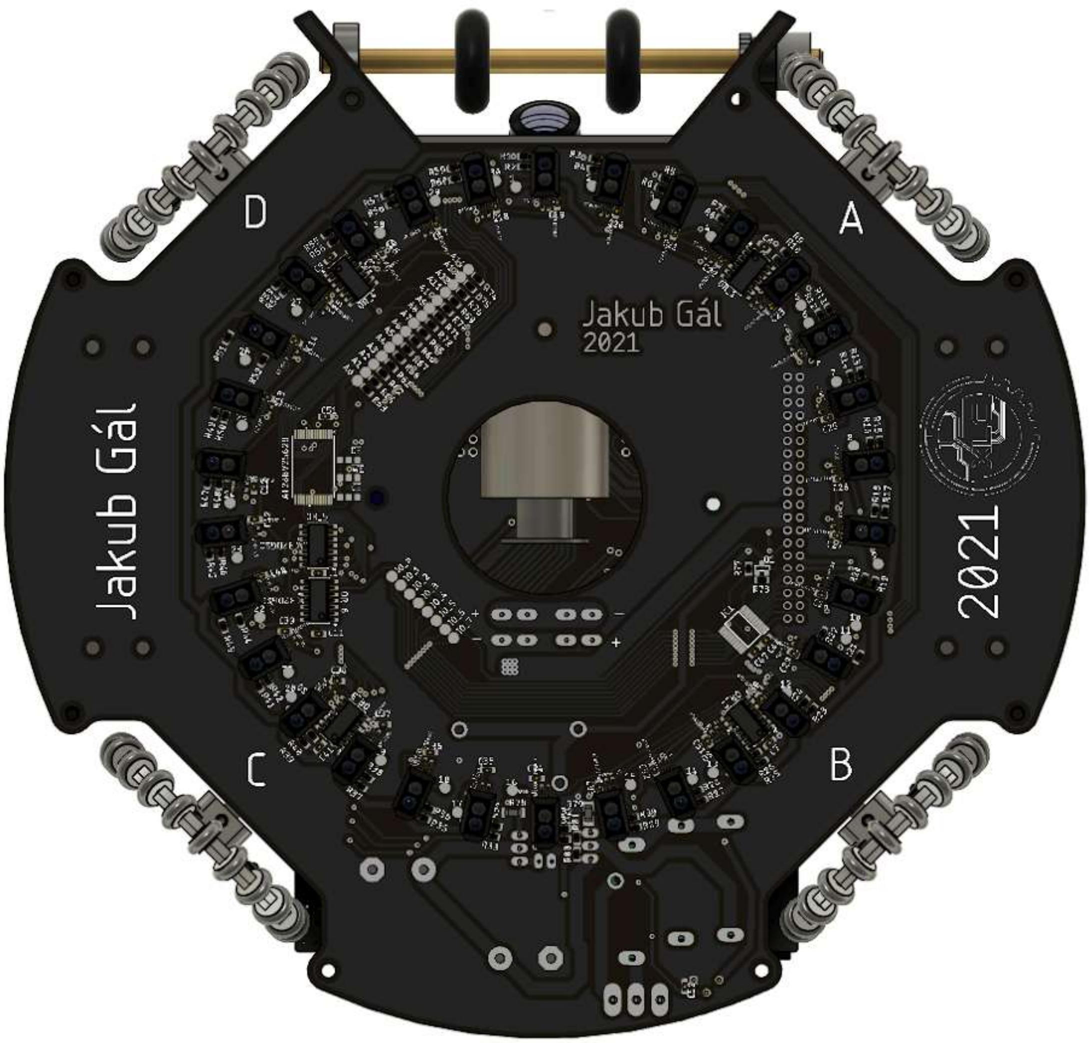
Robot 2 Top View *



Submitted files

Robot 2 Top View - Kushal Sachdeva.HEIC

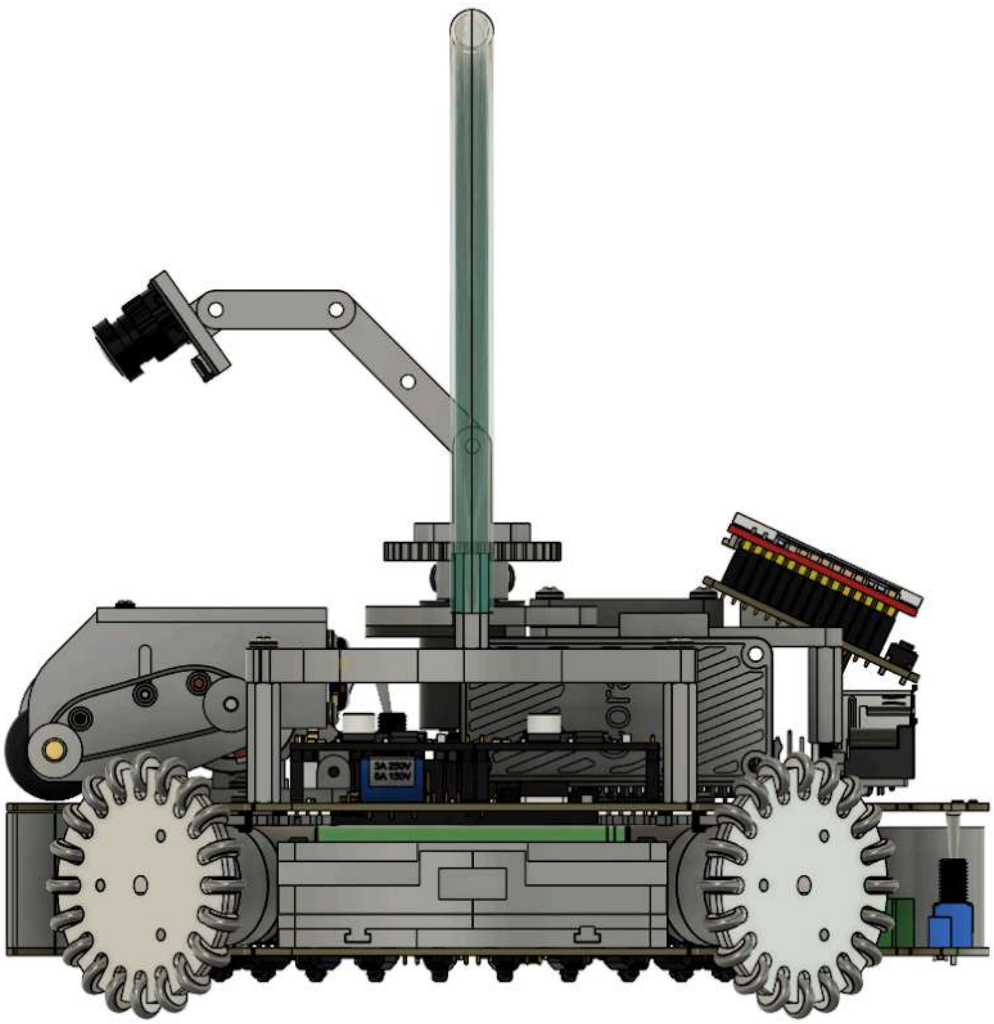
Robot 2 Bottom View *



Submitted files

Robot 2 Bottom View - Kushal Sachdeva.HEIC

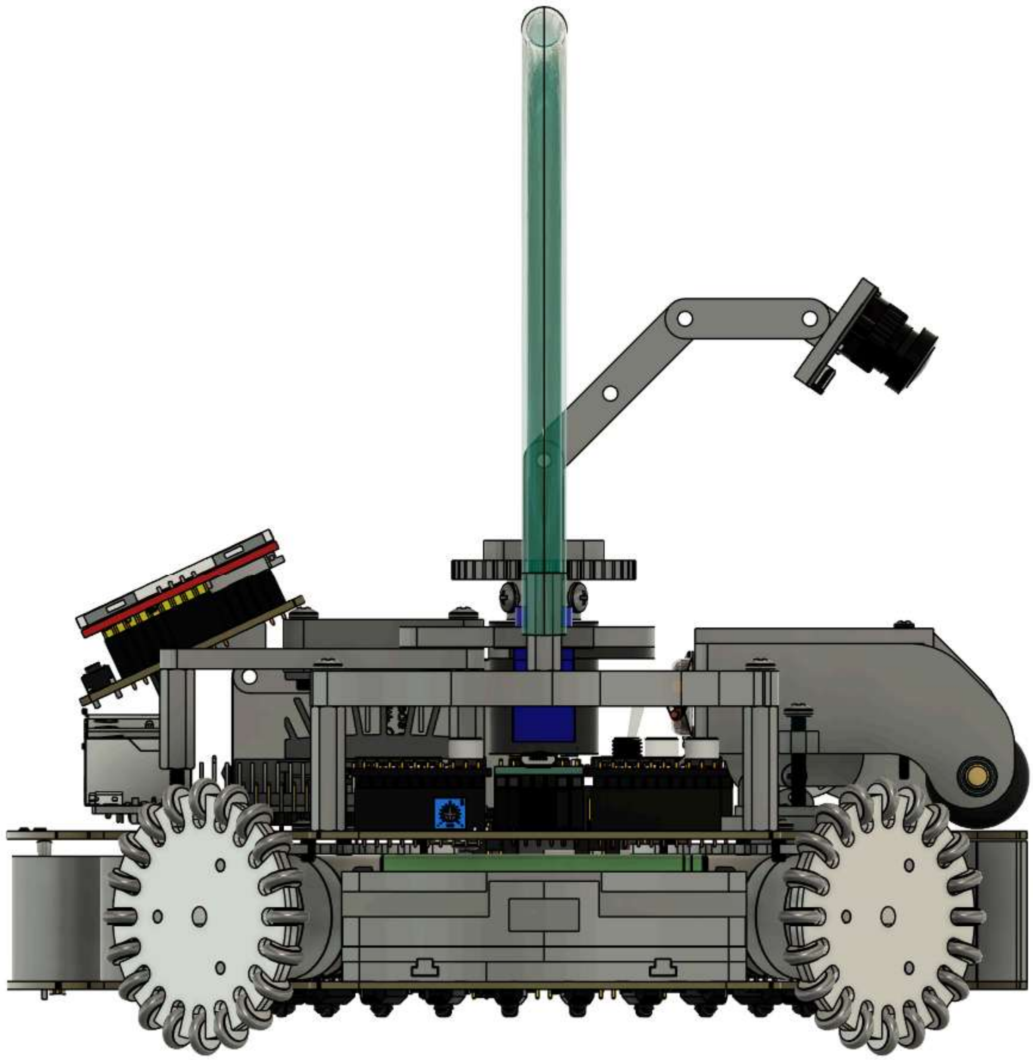
Robot 2 Right View *



Submitted files

Robot 2 Right View - Kushal Sachdeva.HEIC

Robot 2 Left View *



Submitted files

Robot 2 Left View - Kushal Sachdeva.HEIC

How did you design the mechanical parts of your robots? *

Explain which programs you used and how you came up with this design.
Explain what different things you considered to come up with the design.
Explain what you needed to change to make the design better.

Designed in Fusion 360 from hand sketches. With our physics teacher we compared 3- vs 4-omni-wheel layouts, wheel diameters and sensor placement within the 22 cm / 1.5 kg IR-league limits; the 4-wheel diamond won for holonomic agility and even traction. To improve it we added a 3D-printed support-beam frame that keeps the IR ring rigid and planar, sized the capture mouth to the 1.5 cm ball-zone limit, and packed the five PCBs into a compact stack.

How did you build your design? *

Explain which machines you used to build the design.
Explain what services of companies you used to have parts manufactured (e.g. PCBs and mirrors are often made externally)
Explain any changes you needed to make to your design to make it work.

PCBs were made by JLCPCB; parts were 3D-printed on a Bambu Lab printer in PETG (TPU for grip). We tuned print tolerances, reamed the IR-cover holes to 3.2 mm, and reshaped the support beams to clear the Power-PCB terminals and Teensy USB.

How many motors have you used and why? *

This part is for describing
how many motors your team used and why you chose this many motors for
your robots' movement. If you have built your own wheels, it's also
recommended to explain why and how you designed the wheels. Please mention any part numbers of parts you used here as well as in the Bill of Materials (BOM) form.

4 motors, one per omni wheel in a diamond (Pololu 20D mm 12 V gearmotors). Four wheels give true holonomic motion - the robot drives straight at the ball from any bearing while the gyro holds heading - and spread force evenly for pushing duels. We 3D-printed our own omni wheels (PETG hubs + TPU rollers) so the diameter, roller count and grip fit our chassis and the smaller ball.

If your robot has a **kicker**, explain how you designed and built the **mechanics** of the kicker

A solenoid kicker in the capture mouth. The 3D-printed capture area cradles the ball within the 1.5 cm zone; a solenoid plunger strikes the ball forward. A printed mount and coupling align the plunger to the ball centre, and the kick only fires once the capture HC-SR04 confirms the ball is seated.

If your robot has a **dribbler**, explain how you designed and built the **mechanics** of the dribbler.

N/A

CAD design files

Upload all the design files you have of your robot to a GrabCad/GitHub repo and put the link down here. the link must be accessible for anyone when you submit this form. The link will not going to be shared with other teams before the competitions and it's **not mandatory** to provide these files. However, based on the rubrics, you will get **extra points for sharing design files.**

https://github.com/Kushal-Sachdeva78/VVS-Ballers-RoboCup/tree/main/RoboCup%20Internationals%202025-26/CAD

Mechanical Innovation *

Think about the parts of your robot's mechanical system that you are most proud of and try to explain what innovations you came up with that makes you proud. Explain those innovations with as much details as you want.

Our proudest part is the fully self-designed, 3D-printed omni wheels. No off-the-shelf wheel fit our size limit or the smaller 2026 ball, so we built our own: PETG hubs with TPU-gripped rollers, refined over three print rounds to dial in roller diameter, count, grip and concentricity for smooth, slip-free holonomic motion.
Second: a printed support-beam structure that ties the Main, IR-cover and ultrasonic boards into one rigid stack so the IR ring stays planar and aligned through collisions.

Photos of your mechanical designs highlights

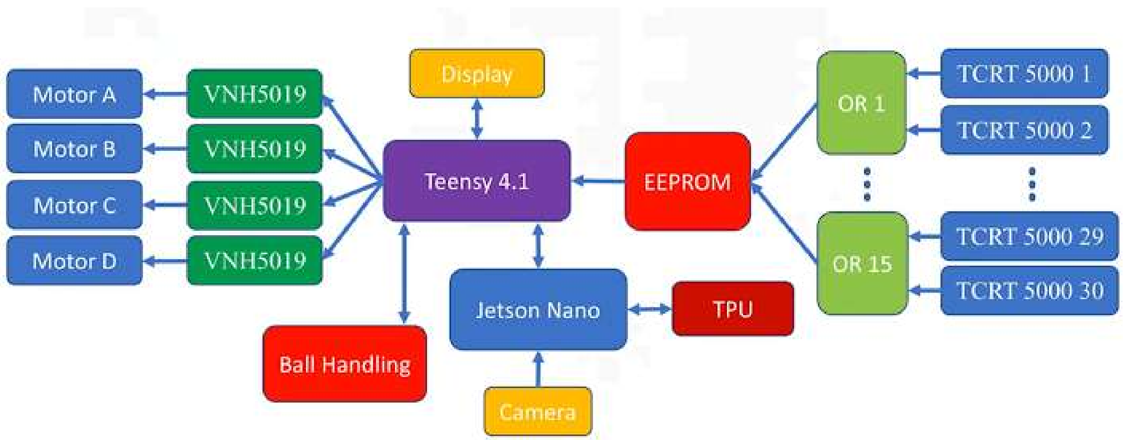
Add up to 5 photos from your mechanical design that you are most proud of. It can be a CAD design screenshot, or a real photo from the finished part.

Submitted files

- Support Beams 3D Model - Kushal Sachdeva.png
- Fully Assembled Robot - Kushal Sachdeva.png
- Mechanical 3D Printed Wheel Opened Up - Kushal Sachdeva.HEIC
- Mechanical 3D Printed Wheel - Kushal Sachdeva.HEIC
- Mechanical door flap for debugging - Kushal Sachdeva.HEIC

Provide us with a block diagram of your robot's electronics

This part is like a whiteboard drawing that shows others how your electronics are built. You don't need to go too much into details. Imagine you are drawing this on a whiteboard to explain to a friend what different electronics parts your robot has. The following photo is an example of what you need to make.



Submitted files

VVS_Ballers_Electronics_Block_Diagram - Kushal Sachdeva.png

How does your power circuits work? *

What voltage levels does your robot use and how do you create them (Linear regulators, buck/boost converters etc...)
Example: Our robot has a 14.4V battery pack that is regulated to 5V for our Arduino and used directly by the motor drivers.

A 12 V LiPo pack feeds a Power PCB which generates: 5 V (buck) for the Teensy boards boards and the OpenMV camera; 3.3 V (buck) for sensor logic; and 48 V (boost) for the solenoid kicker (All this is done on one PCB). The 12 V rail goes directly to the motor drivers. Bulk capacitors sit on each rail

How do you drive your motors? Explain the circuits you use for that *

Each omni wheel has its own DRV8263H brushed driver (12 V power, 3.3 V logic, PH/EN mode); the Main Teensy feeds each a direction + PWM pin. Each driver's IPROPI current output goes to an Arduino Nano that pulls DRVOFF to coast a motor over ~1 A (stall ~1.6 A), preventing stall overheating.

What kind of micro controller or board do you use for your robot? Why did you decide to use this part for your robot? If you have more than 1 processor, explain each one separately.
*

3× Teensy 4.1:
(1) Main board: chosen for its 600 MHz speed and many hardware UARTs (it must service 5 serial links + the comms module at 100 Hz).
(2) IR ring board: needs to read 16 analog channels fast with on-chip averaging;
(3) Line ring board: reads 18 analog channels at >500 Hz. 2×
Arduino Nano Every:
(4) Ultrasonic board: Receives values from 4 HC-SR04s;
(5) Motor-current limiter: Current protection runs independently of the main code.
Plus the OpenMV H7 as a vision coprocessor.

How do you use your processor to move your motors? *

The main Teensy 4.1 PWMs four DRV8263H drivers (one direction pin + one analogWrite PWM pin each, shared SLEEP, 12 V). It mixes a translation pattern with a gyro heading-correction term, so it strafes any direction while facing forward. A separate Nano caps current via IPROPI/DRVOFF.

How does your ball detection sensors and/or camera[s] work? *

A 16-channel TSSP58038 IR receiver ring is the primary ball sensor - the IR-league ball emits IR, and each receiver's reading becomes an intensity (baseline - raw). An OpenMV H7 camera complements it: besides finding the goal, keeper and open-corner aim, it also detects the orange ball, letting us cross-check and reconfirm the IR ring's bearing - camera+IR fusion that helps at short range where the IR ring alone struggled. The camera sends its data to the main board as a 11-byte frame.

How does your line detection circuits work? *

A dedicated Line PCB carries four QTR-MD-05A IR-reflectance arrays - 18 sensors facing the floor on all sides. The white field line reflects differently from the green carpet (white reads low), and a board flags 'line' when >=2 of its channels cross threshold. The four ultrasonics range the walls.

What sensors do you use for navigation and how are these sensors connected to your processor? What sensors do you use to find your position in the field? What about the direction your robot faces? *

Direction: a BNO055 9-DOF IMU on the main board (I2C/Wire2, address 0x28) in IMUPLUS mode, so the drifting magnetometer is ignored and gyro+accel give a stable heading for the

100 Hz hold-PID. Position: four HC-SR04 ultrasonics, one per side, wired to the ultrasonic Nano that streams their distances to the main board for boundary and goal standoff.

How do you drive your kicker system? How does the circuit make the kicker work? *

A relay module (IN1 = pin 22, active-HIGH) switches the 48 V solenoid; a capture HC-SR04 over the mouth (TRIG 20 / ECHO 21) confirms the ball before firing. Firmware safety: 2-sample capture debounce, a fixed 500 ms pulse, 1.5 s cooldown, a hard 600 ms max-on cutoff, and a boot-safe relay state.

How does your dribbler system work? What components and circuits did you use to drive it?

N/A

Schematics of your robot

If you designed schematics (circuit diagrams) for your robot, upload a PDF or picture of the schematics here

Not mandatory but helps community to grow and will get you some extra points for documentation that counts into final score.

Submitted files

- Line Schematic - Kushal Sachdeva.png
- Main 2.0 Schematic - Kushal Sachdeva.png
- Ultrasonic Schematic - Kushal Sachdeva.png
- IR Schematic - Kushal Sachdeva.png
- Power Schematic - Kushal Sachdeva.png

PCB of your robot

If you designed a PCB for your robot, upload a PDF or picture of the PCB here

Not mandatory but helps community to grow and will get you some extra points for documentation that counts into final score.

Submitted files

- Power PCB - Kushal Sachdeva.png
- IR PCB - Kushal Sachdeva.png
- Ultrasonic PCB - Kushal Sachdeva.png
- Main PCB 2.0 - Kushal Sachdeva.png
- Line PCB - Kushal Sachdeva.png

Electronics Innovations *

What part of your electronics are you most proud of? Explain these parts in details and explore any innovation you came up with make this designs work

Our federated multi-PCB architecture: three purpose-built boards talk over CRC-8-framed UART, so a glitch on any wire is rejected, not acted on. We are proudest of three boards. The Power PCB is compact yet generates and distributes four rails (12 V, 5 V, 3.3 V, 48 V) cleanly and continuously across the whole robot from one battery. The IR PCB gives each of its 16 receivers its own RC filter, smoothing the pulsed IR into a clean analog level the Teensy reads as a precise bearing. The Main PCB integrates everything on one board - it takes data from every sensor link, runs the heading PID, drives all four motors, and carries the BNO055 IMU and the solenoid kicker at once.

Photo of your circuit boards highlights *

Add up to 5 photos from your finished and soldered circuits that you are most proud of. It can be a CAD design screenshot, or a real photo from the finished part.

Submitted files

- Power PCB - Kushal Sachdeva.HEIC
- Power PCB 3D model - Kushal Sachdeva.png

5V, 3.3V & UART Wiring - Kushal Sachdeva.HEIC

IR PCB RC Circuit - Kushal Sachdeva.HEIC

Arduino Nano Current Controller - Kushal Sachdeva.HEIC

How do you find where the ball is? How do you read the data from the ball detection sensors and/or camera? *

The IR board reads all 16 channels, converts each to intensity = baseline - raw, rejects ambient and damps lone spikes. It then takes an intensity-weighted vector sum (atan2 over all lit sensors) for a smooth bearing, with hysteresis (60/35) and EWMA smoothing, and sends <dir>a<dist>b (500 = no ball) to the main board. The camera is read as an 11-byte frame; stale links (>200 ms) are ignored.

How does your algorithm work to catch the ball? Is there a difference between your robots in how they move towards the ball? Explain the differences. *

Attacker: it sorts the IR bearing into a drive sector and translates straight at the ball (speed 250) while the gyro holds heading; on capture it switches to aim-and-kick.
Defender: it does not chase up-field - it stays on its goal line and runs a lateral PID on the ball's bearing to keep its body between ball and goal, with an intercept boost when the bearing moves fast, clearing on contact.

How do you use your sensors in your algorithm to find your position inside the field and how do you use that position to move your robots around? *

The four ultrasonics are used reactively (no x/y estimate), each giving its wall distance. The attacker uses them with the line ring to flee field edges; the defender centres in the mouth from the left/right difference and holds a set distance-range from its own goal via the back sensor. Drive is holonomic: a translation vector plus the heading-PID correction, normalised so no wheel saturates.

How does your robot find the lines to stay inside the field? What algorithms do you use to avoid going out of bounds? *

Walls are sensed by ultrasonics, the white boundary by four QTR-MD-05A reflectance arrays (>=2 white channels = line). Attacker: a side is the edge only when its ultrasonic reads <300 mm AND that side sees white, debounced 3 passes; it then flees, never on a stale link. This two-of-two rule stops a false reading triggering an out-of-bounds penalty. Defender: the line keeps it on the goal edge.

What algorithms do you use to score goals? How do you use your kicker and dribbler to handle the ball? *

The camera finds the goal and keeper and returns the open-corner bearing. On capture (mouth HC-SR04 <45 mm, 2 samples) the attacker holds heading, then spins in place until the open corner is within 5 deg ahead, and only then fires the 48 V solenoid (500 ms pulse, 1.5 s cooldown, 600 ms hard cutoff). The defender uses the same kicker for a short clearing kick. No dribbler.

What algorithms do you use to avoid the opponent team scoring? How do your robots defend your own goal? *

The defender's Defender_Full runs GUARD (centre on the line, hold standoff) → TRACK (lateral-PID onto the ball, intercept boost) → CLEAR (push/kick the ball away on contact) → RECOVER (re-home to the line). A back-wall safety rule forbids reversing toward its own goal when the rear wall is < 90 mm, so a bad reading can't drive it into its own net.

Do your robots communicate with each other? How do you use this communication to your advantage?

Yes - the robots share status over a 2.4 GHz link (the only band RCJ permits for inter-robot comms). We use it for automatic role-switching: if one robot drops off the link (fouled, penalised or removed), its partner switches from attacker to defender so our goal is never left open while we are a robot down; they resume normal roles when it returns at kick-off.

Software Innovations *

Tell us about any specific code or algorithm you build that you are the most proud of. Explain how you came up with this innovation and how it helps you win more games.

Both are about clean data from cheap sensors.
(1) Ultrasonic smoothing: we ping opposite sensors together to kill cross-talk, temperature-compensate distance, run a per-sensor median-of-3, and coast over dropouts so a missed echo never blinks a wall away.

(2) IR tracking: instead of strongest-sensor, we sum intensity-weighted vectors over all lit receivers (atan2) with hysteresis and spike damping, for a smooth bearing on the small fast ball. Both feed our fused line+ultrasonic boundary escape

GitHub link

Give us a link to your GitHub repo containing all your firmware/software.
Consider making one if you don't have one.

(Based on the rubrics document, you can get extra points if you publish your code)

https://github.com/Kushal-Sachdeva78/VVS-Ballers-RoboCup/tree/main/RoboCup%20Internationals%202025-26/Firmware

Bill of Materials (BOM) *

List all **the main parts** (e.g. your processor, motors, wheels, sensors, cameras, major electronics and mechanical parts) used in your robots with their count and unit price

Use the following template for the parts:

https://docs.google.com/spreadsheets/d/1DsPMRiU3CD8XgNf5UaV88hU5cOljiG35LtWss3hpuA4

Submitted files

BOM_VVS Ballers - Kushal Sachdeva

How much did it cost you to build your robots? *

Please provide three numbers, one for the final components which went into your robot, one for money you spend for trial and error building your robots and finally the cost of getting the environment ready to build your robots (building the field, the carpet, soldering iron, oscilloscope, etc). Please provide the currency you used in your calculations and the exchange rate to US Dollars at the time of filling this form.

Example:
Robots (cost of components that are in your robots right now): 3000 Euro each
Experiments (failed builds, broken hardware etc.): 2000 Euro
Environment (fields, balls, etc.) : 1000 Euro
1 Euro= 1.14 USD

Robots (cost of components that are in your robots right now): 600 Euro each
Experiments (failed builds, broken hardware etc.): 1200 Euro

Spare Parts (spare motor drivers, sensors, fuses, PCBs, etc.): 1000 Euro
Environment (fields, balls, 3D printer, soldering station, power tools, etc.) : 1100 Euro

How did you gathered the funds to build the robots? *

Example:
30% sponsors
20% school
50% parents

50% Sponsors
0% School
50% Parents

How affordable was it to compete in RoboCupJunior Soccer? *

	1	2	3	4	5	6	7	8	9	10	
Very Expensive	○	○	○	◉	○	○	○	○	○	○	Very Affordable

Have you checked all of your answers? *

◉ Yes!

We publish TDPs and posters during or after the competition as described in the beginning
*

◉ Yes, we acknowledge everything submitted in the above form can be published.